

Arithmetic & Number Sense



Order of operations

- 1 Evaluate $12 + 6 \times (9 - 5) \div 3$.
 - 2 Evaluate $8 + 2 \times 5^2 - 30 \div 6$.
 - 3 Evaluate $\frac{18 - 3 \times 4}{2} + 7$.
 - 4 Evaluate $(7 - 2)^2 - 3 \times (6 - 4)$.
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Powers and roots

- 5 Evaluate 3^4 .
 - 6 Evaluate $\sqrt{144}$.
 - 7 Evaluate $2^3 \times 2^2$ using the laws of exponents.
 - 8 Evaluate $\sqrt{81} + \sqrt{16}$.
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Working with fractions

- 9 Compute $\frac{3}{4} + \frac{5}{6}$.
 - 10 Compute $\frac{7}{8} - \frac{1}{3}$.
 - 11 Compute $\frac{2}{3} \times \frac{9}{10}$.
 - 12 Compute $\frac{5}{6} \div \frac{5}{9}$.
 - 13 Which is larger, $\frac{5}{7}$ or $\frac{4}{5}$?
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Decimals

- 14 Compute $3.45 + 2.7$.
- 15 Compute $8.6 - 3.25$.

16 Compute 1.2×0.5 .

17 Compute $7.5 \div 0.25$.

Place value and order of magnitude

18 Write 4,500,000 in the form $a \times 10^n$ with $1 \leq a < 10$.

19 Which is greater, 6×10^4 or 58,000?

20 Order from least to greatest: 0.6, $\frac{2}{3}$, 0.65.

Mixed numbers and improper fractions

21 Convert $4\frac{2}{5}$ to an improper fraction.

22 Convert $\frac{47}{6}$ to a mixed number.

23 Compute $2\frac{1}{3} + 1\frac{3}{4}$.

24 Compute $5\frac{1}{2} - 2\frac{2}{3}$.

Absolute value and signed numbers

25 Evaluate $|-7| + |-3|$.

26 Evaluate $-8 + 5 \times (-2)$.

27 Evaluate $|4 - 9| - |2 - 6|$.

Factors, multiples, GCD and LCM

28 Find the greatest common divisor (GCD) of 48 and 60.

29 Find the least common multiple (LCM) of 8 and 12.

30 List all the factors of 36, and state how many there are.

- 31 Is 91 a prime number? Justify your answer.
- 32 Find the greatest common factor of 84 and 126.
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Divisibility rules

- 33 Without dividing, determine whether 4,578 is divisible by 3.
- 34 Is 2,340 divisible by 9?
- 35 Is 715 divisible by 5?
- 36 Is 1,244 divisible by 4?
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Estimation and rounding

- 37 Estimate 198×51 by rounding each number to the nearest ten, then find the exact product.
- 38 Round 2.4863 to two decimal places.
- 39 Estimate $\sqrt{50}$ to the nearest whole number.
- 40 A shipment weighs approximately 2,850 kg. Round this to the nearest hundred kilograms.
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Number properties

- 41 If n is an even integer, is $n^2 + n$ always even, always odd, or can it be either?
- 42 Find the remainder when 47 is divided by 6.
- 43 Is 637 divisible by 7? Justify your answer.
- 44 List all prime numbers between 30 and 50.
- 45 If the sum of two integers is odd, is their product always even, always odd, or can it be either?
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Word problems

- 46** A school orders 158 notebooks to be shared as equally as possible among 12 classrooms. How many notebooks does each classroom receive, and how many are left over?
- 47** A caterer must seat 250 guests at tables that each hold 8 people. What is the minimum number of tables required?
- 48** Three friends share a restaurant bill of 186 riyals equally, then each adds a 10-riyal tip. How much does each person pay in total?
- 49** A red light flashes every 8 seconds and a green light flashes every 12 seconds. If both flash together right now, after how many seconds will they next flash together at the same time?
- 50** A water tank holds 480 liters and is filled by a pump running at 15 liters per minute. How long does it take to fill the tank from empty?
- 51** A charity collects 2,340 riyals from 45 equal donations. How much was each donation?